For a, b real numbers, b nonzero, we write

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

for $a = r\cos\phi$, $b = r\sin\phi$, the eigenvalues of the matrix on the left being $a \pm bi$, always nonreal since b is given to be nonzero.

Can find r and ϕ via:

$$r^2 = a^2 + b^2 \qquad \text{and} \qquad \tan\phi = \frac{b}{a}$$

Just be careful about what sign to choose for r and which quadrant you pick ϕ to be in, need a and b to have the appropriate signs. Could be multiple options, just be sure the one you take does match accordingly.

One reason to do this is (k some positive integer):

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}^k = \left(\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \right)^k$$

How to multiply out the right side though? Well matrix multiplication is not commutative so we might be wary. But key is

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} = rI$$

So then that will be commutative since I commutes with every matrix, and r is just a scalar. Can even just think of r being a scalar in front of the second matrix, not a part of its own matrix. Either way, it becomes:

$$r^k \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}^k$$

And then the matrix remaining is a rotation by ϕ so applying it k times, will rotate by $k\phi$ giving this to be

$$r^k \begin{bmatrix} \cos\ k\phi & -\sin\ k\phi \\ \sin\ k\phi & \cos\ k\phi \end{bmatrix}$$

Can also prove this via DeMoivre's Theorem type ideas:

$$[\,r\cos\phi + i\sin\phi\,]^k = r^k[\cos k\phi + i\sin k\phi\,]$$

So key is

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}^k$$

can be computed fairly conveniently, not dissimilar to diagonalization. Just we can't diagonalize this over $\mathbb{R}$ so this is a slightly different way of accomplishing that.

So this is why it might be useful to show any 2 x 2 real matrix with nonreal eigenvalues is similar to matrix of type

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

If we want the high power of such a matrix, since it can't be diagonalized over $\mathbb{R}$ this is a replacement option, sort of a "pseudo-diagonalization" which breaks down the matrix and allows us, via the earlier ideas, to compute say a high power of it if that's what we want.

So how to do that, plus finding eigenvectors of such a matrix to begin with? Bit different with imaginary numbers involved.

5.5.3

$$A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = (1 - \lambda)(3 - \lambda) + 10 \stackrel{?}{=} 0$$

----> $$\lambda^2 - 4\lambda + 13 = 0$$

----> $$(\lambda - 2)^2 + 3^2 = 0$$

----> $$\lambda = 2 \pm 3i$$

For $\lambda = 2 + 3i$ find eigenvalue v:

$$A - (2 + 3i)I : \begin{bmatrix} -1 - 3i & 5 \\ -2 & 1 - 3i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} (-1 - 3i)x_1 + 5x_2 = 0 \\ -2x_1 + (1 - 3i)x_2 = 0 \end{cases}$$

Key is these two equations are equivalent - they have to be even though they might look different to naked eye.

Why? Because if they weren't we'd have two independent equations in two variables. Ok a bit different due to nonreal coefficients, but same ideas would then force this to have only trivial solution. And that would contradict this being an eigenvalue.

So we can just use second equation, plug in some convenient nonzero value for one variable, then find the other:

$$x_2 = 2 \text{ ----> } x_1 = 1 - 3i$$

So $v = \begin{bmatrix} 1 - 3i \\ 2 \end{bmatrix}$ is an eigenvector for $\lambda = 2 + 3i$

Remember any nonzero constant multiple of an eigenvector is still an eigenvector.

And that includes a nonreal constant multiple. So if you do this calculation a bit differently and get a different looking eigenvector, that doesn't necessarily mean you did anything wrong, might just be a constant multiple.

And this is even more amplified with nonreal numbers mixed in as it's less obvious when two nonreal vectors are multiples of each other, the same reason you might have been apprehensive about the two equations being multiples of each other as with the i mixed in, it's less perceptible that that's the case.

What about the other eigenvalue, 2 - 3i? Turns out we don't need the eigenvector for that, and even if we did we wouldn't need to do additional computation to find it.

Why? First recall notation that for z any complex number that $\bar{z}$ is the <u>complex conjugate</u> of z. That is,

$$z = a + bi \quad \dashrightarrow \quad \bar{z} = a - bi$$

It turns out for any real square matrix A with a nonreal eigenvalue λ that $\bar{\lambda}$ is also an eigenvalue of A and if v is an eigenvector for λ, then $\bar{v}$ is an eigenvector for $\bar{\lambda}$. Why?

$$Av = \lambda v$$

$$\dashrightarrow \quad \overline{Av} = \overline{\lambda v}$$

$$\dashrightarrow \quad \bar{A}\bar{v} = \bar{\lambda}\bar{v}$$

since conjugate of product is product of conjugates. This then gives

$$A\bar{v} = \bar{\lambda}\bar{v}$$

because A is real, so it equals its conjugate. And this indeed proves $\bar{\lambda}$ is also an eigenvalue for A with $\bar{v}$ being an eigenvector for it.

The computation in this past question indirectly shows another reason why we don't really want to deal with nonreal eigenvalues for 3 x 3 and larger.

Main point is the trick of the two equations being the same so just plugging in a value for one of them to find the eigenvector won't work with three or more equations.

Would have to do more (row reduction or variable elimination) - both obviously harder with i mixed in.

Say also there is an eigenspace of dimension higher than 1 for some nonreal eigenvalue so that we need at least two linearly independent eigenvectors for that eigenvalue. What would that mean though with i mixed in? Trickier for sure!

Fully diagonalizing a 3 x 3 real matrix even in the "easiest" case of all eigenvalues being distinct real numbers is still quite lengthy. Mixing in nonreal eigenvalues will generally just make it too extreme. Doable of course still though.

In general for A real 2 x 2 matrix with nonreal eigenvalues $a \pm bi$, say v is an eigenvector for $a - bi$, perhaps

$$v = \begin{bmatrix} j_1 + k_1 i \\ j_2 + k_2 i \end{bmatrix}$$

Then we can write

$$A = PCP^{-1}$$

where $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $P = \begin{bmatrix} j_1 & k_1 \\ j_2 & k_2 \end{bmatrix}$

Why? Well first column of AP is $A \begin{bmatrix} j_1 \\ j_2 \end{bmatrix}$ while

for PC it is

$$\begin{bmatrix} j_1 a + k_1 b \\ j_2 a + k_2 b \end{bmatrix}$$

Now for the stated λ we have $Av = \lambda v$
for which the right side of this is equal to:

$$(a - bi) \begin{bmatrix} j_1 + k_1 i \\ j_2 + k_2 i \end{bmatrix}$$

so the real part of this is:

$$\begin{bmatrix} j_1 a + k_1 b \\ j_2 a + k_2 b \end{bmatrix}$$

while for Av the real part is:

$$A \begin{bmatrix} j_1 \\ j_2 \end{bmatrix}$$

And these are exactly the first columns in AP, PC that we mentioned just before.

And since those real parts must match, this proves the first columns in AP, PC match as well. Likewise imaginary parts will give second columns to match, proving this works overall, as desired.

5.5.15

$$A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

Found $\lambda = 2 + 3i$, $v = \begin{bmatrix} 1 - 3i \\ 2 \end{bmatrix}$ in 5.5.3

So we have:

$$C = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & -3 \\ 2 & 0 \end{bmatrix}$$

How might we use this?

E.g. $$A^{100} = PC^{100}P^{-1}$$

And to compute C^{100}, write C as product of scaling and rotation matrix.

There is often confusion in this setting about the sign of the top right term in the matrix C. We say for the eigenvalue $a - bi$, top right is $-b$, but if we used the eigenvalue $a + bi$?

You can use either eigenvalue, key is whichever one you use, the top right entry of C should match the imaginary part of that eigenvalue.

So it can work both ways, point being if you switch the eigenvalue to the conjugate, that will also switch the eigenvector to its conjugate. The matrix P will compensate is the point.

So we could also do:

$$C = \begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix} \quad , \quad P = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}$$

via taking the other eigenvalue and conjugating the eigenvector (shown earlier) which alters the sign of the entries in the second column of P.

Just like diagonalization, there are multiple forms, done properly, it can work in slightly different ways.

Chapter 6

Chapters 6 + 7, the culmination of the linear algebra portion of the class, are perhaps the hardest of the class. Ok, as always somewhat matter of opinion.

Chapter 7 is a short chapter, largely similar type ideas to Chapter 6 (but also some ideas from Chapter 5), hence why I'm sort of grouping them together.

Chapters 5 and 6 are both tough since they are quite cumulative in nature, both hinging on ideas learned earlier in the class.

But Chapter 5 is mostly ideas with solving homogeneous equations, determinants, stuff from Chapters 1 - 3, which most have an easier time with.

Chapter 6 is building on ideas from vector spaces, stuff from Chapter 4 largely which was tougher for

most. Hence why I think most find Chapter 6 harder.

6.1 Inner Product and Orthogonality

The concept of inner product is quite important in higher level math. Eventually (end of this chapter) we will talk about special type of vector space, which is called inner product space.

But ok, that's not focus for now, mentioned more since we start off dealing with a special type of inner product.

So that is a good way to get an early introduction to the concept.

Take $u, v \in \mathbb{R}^n$, denoted by

$$u = (u_1, u_2, \ldots, u_n) \quad , \quad v = (v_1, v_2, \ldots, v_n)$$

Define dot product of u and v as:

$$u \cdot v = (u_1, u_2, \ldots, u_n) \cdot (v_1, v_2, \ldots, v_n)$$

dot product

$$= u_1 v_1 + u_2 v_2 + \ldots + u_n v_n$$

Aside: We often write vectors horizontally rather than vertically here. Horizontally is really more typical actually, even though we more did it vertically earlier the class as it needed to be that way when doing matrix multiplication as we were.

But understand no fundamental difference. If question asks you to solve for such and such vector, it would be perfectly fine to write the answer as a vertical vector or a horizontal one.

Dot product is one type of inner product, definitely the most used one in this class.

We already used this a lot even though didn't mention it by name much. In matrix multiplication when doing row "times" column, that really was the dot product of the two. Not really traditional multiplication of the two (how could it be since those aren't real numbers). But ok traditional multiplication is obviously involved.

Required Properties:

These apply to all inner products (hence why this is a good introduction to those). But ok will use the notation for dot product here since that's what we will be working with for the large part of this chapter.

But they would work whether you say $u \cdot v$ for dot product or $\langle u, v \rangle$ for some generic inner product.

These properties need to hold for all u, v in the space.

(1) Commutative. That is, $u \cdot v = v \cdot u$

(2) Distributive law with addition. Which means

$$u \cdot (v + w) = u \cdot v + u \cdot w$$

(3) Scalar is "movable". For any such c, $(cu) \cdot v = u \cdot (cv)$

(4) $u \cdot u \geq 0$ with $u \cdot u = 0$ only when $u = 0$

Where did we see $u \cdot u$ earlier? When doing $A^T A$ earlier for any matrix A. The diagonal entries of that are the columns of A dotted with themselves. Hence why they are ≥ 0 (in Midterm 1 Review)

This relationship is part of the reason why transpose really becomes important in this chapter.

Several other relevant ideas involving these concepts:

Norm (length) of vector is:

$$\| v \| = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2}$$

Equivalently,

$$\| v \|^2 = v \cdot v$$

Distance between u and v is: $\| u - v \|$

Very important concept in this chapter: two vectors u, v are <u>orthogonal</u> if $u \cdot v = 0$

When in $\mathbb{R}^2$ or $\mathbb{R}^3$ this can be thought of those vectors being perpendicular, i.e intersecting at a right angle. We might not be inclined to think that way in higher dimensions, but dot product being 0 is not different in terms of mathematically testing the dot product, no matter what dimension we are in.

So orthogonal can be viewed as a generalization of the term perpendicular.

A couple other ways we can look at this:

u and v are orthogonal exactly when

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2$$

This is Pythagoras based idea so natural. Why is this?

$$\begin{aligned}\|u+v\|^2 &= (u+v)\cdot(u+v)\\ &= u\cdot u + u\cdot v + v\cdot u + v\cdot v\\ &= \|u\|^2 + \|v\|^2 + 2(u\cdot v)\end{aligned}$$

so last term being 0 versus not is the key, as desired.

Another geometric idea which uses dot product is

$$u\cdot v = \|u\|\,\|v\|\cos\theta$$

where θ is the angle between u and v.

This is based on Law of Cosines and yet another reason why dot product being 0 relates to perpendicular.

So that's how we test if two vectors are orthogonal.

A much stronger thing: if W is subspace of $\mathbb{R}^m$ can we have a vector u which is "orthogonal" to W?

First off, what does that even mean?

Again we defined what it means for vector u to be orthogonal to another vector v. But what does it mean for u to be orthogonal to a subspace W?

It means u is orthogonal to *every* vector in W. Now this seems very strong at first. One thing for a vector to be orthogonal to one single other one, but a subspace is usually infinitely many vectors seems it that would not even be possible perhaps.

But in higher dimensions and the fact that W is a subspace (very orderly subset), this is usually possible.

<u>Ex:</u> In $\mathbb{R}^3$ take W to be the xy-plane. Is the z-axis orthogonal to W?

Pictorially or via cross product you know it is. But mathematically a vector in W is of form $(x, y, 0)$ while z-axis is represented with $(0, 0, 1)$. And regardless of values of x, y we do always have

$$(x, y, 0) \cdot (0, 0, 1) = 0$$

So the z-axis is orthogonal to W here.

In general the set of vectors orthogonal to any subspace W is the <u>orthogonal complement</u> of W, denoted as $W^{\perp}$

One good place we use this actually gives us another interesting way to look at/understand the typical Row A, Col A, Nul A. That is, for any real matrix A we have:

$$(\text{Row } A)^{\perp} = \text{Nul } A$$

Why is this true?

Well Nul A means all solutions to $Ax = 0$. Vector x satisfying this has to have dot product 0 with every row, that's how matrix multiplication works after all.

Thus must be in orthogonal complement of row space.

So that basically proves the result.

Then similarly:

$$(\text{Col } A)^{\perp} = \text{Nul } (A^T)$$

Back in Chapter 4 at various times we asked questions about creating matrices which had same row space as null space, same column space as null space, and other things relating these to each other.

Well as this shows, the potential relationships between those might have been deeper than we originally thought - could have helped us answer some of those questions more easily perhaps.

But that's not the key point, big issue is orthogonality is very important and these relationships help both ways, helps us understand those subspaces better and what we know of those subspaces already (e.g. rank-nullity) helps us understand orthogonal complement better.

These ideas are a big part of this chapter as we will over the next few classes.